HitPredict: Predicting Billboard Hits Using Spotify Data

UE24CS352A Machine Learning Mini-Project | Team ID: A-06 | Project No: 21
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1. Problem Statement

Can a song's Spotify audio features tell us whether it is a Billboard Hot 100 song? We cast this as binary classification (**1 = Hot 100 hit, 0 = non-hit**) and compare Logistic Regression, Decision Tree and Random Forest. The task follows *HitPredict* (Georgieva, Suta & Burton, Stanford CS229, 2018), which reached about 75% validation accuracy on a balanced set of ~4,000 songs using Spotify features plus an artist-history score.

2. Dataset Details

Two public Kaggle datasets were merged into **7,713 songs × 14 features**. **Hits (6,879):** *Billboard_Hot100_Songs_Spotify_1946-2022* (label 1). **Non-hits (834):** songs from *playlist_2010to2023* whose normalised title does not appear in the Billboard file and whose year is ≤ 2022 (label 0). Features: Popularity, Acousticness, Danceability, Energy, Instrumentalness, Liveness, Loudness, Speechiness, Tempo, Valence, Key, Mode, Time Signature, Song Length. No values are missing. Two properties of the data shape every result:

- **Heavy imbalance:** 89.2% of rows are hits. Predicting “hit” for every song already scores 89.2% accuracy.
- **Duplicates:** 358 exact duplicate rows (songs charting in several years) were found and removed, leaving 7,355 unique songs (817 non-hits), so copies cannot sit in both train and test.

3. Approach

Preprocessing. Titles are lower-cased and stripped of quotes, hyphens and brackets; Billboard matches are removed from the playlist data; common features are renamed and concatenated; exact duplicates are dropped; features are standardised (for Logistic Regression). **Models.** Logistic Regression (baseline), Decision Tree (max depth 5) and Random Forest (200 trees, max depth 10), all with `class_weight='balanced'` so errors on the rare non-hit class count more. A majority-class **dummy baseline** is included. **Evaluation.** Stratified 5-fold cross-validation (mean ± std), reporting accuracy, **balanced accuracy**, macro-F1, ROC-AUC and non-hit recall/precision, plus a **Popularity ablation** (all 14 features vs. 13 pure audio features).

4. Implementation Overview

Python (pandas, NumPy, scikit-learn, matplotlib, seaborn) in `HitPredict_v2.ipynb`: load/build data → audit (duplicates, class balance, correlations) → reproduce the original un-weighted pipeline → corrected cross-validated comparison → hold-out confusion matrix and ROC → feature importance → `predict_song()` demo that returns P(hit) for a feature vector. The repository README explains setup and running.

5. Results

The original pipeline was misleading. Un-weighted models scored 88.9–89.2% accuracy, no better than the 89.2% baseline, and identified only 0.6–1.8% of non-hits (balanced accuracy ≈ 0.50). After correcting the evaluation (5-fold CV, all features):

Model	Accuracy	Balanced acc.	Macro-F1	ROC-AUC	Non-hit recall	Non-hit prec.
Majority baseline	0.889	0.500	0.471	0.500	0.000	0.000
Logistic Regression	0.702	0.754	0.592	0.806	0.820	0.247
Decision Tree	0.682	0.746	0.577	0.789	0.829	0.237
Random Forest	0.753	0.739	0.619	0.828	0.721	0.271

Without Popularity, balanced accuracy / ROC-AUC fall to 0.721 / 0.785 (LR), 0.723 / 0.763 (DT) and 0.694 / 0.796 (RF): audio features alone still carry signal, but Popularity adds 0.02–0.03 AUC. Random Forest feature importance ranks **Popularity (0.25)** and **Loudness (0.21)** far above the rest; Mode and Time Signature contribute almost nothing.

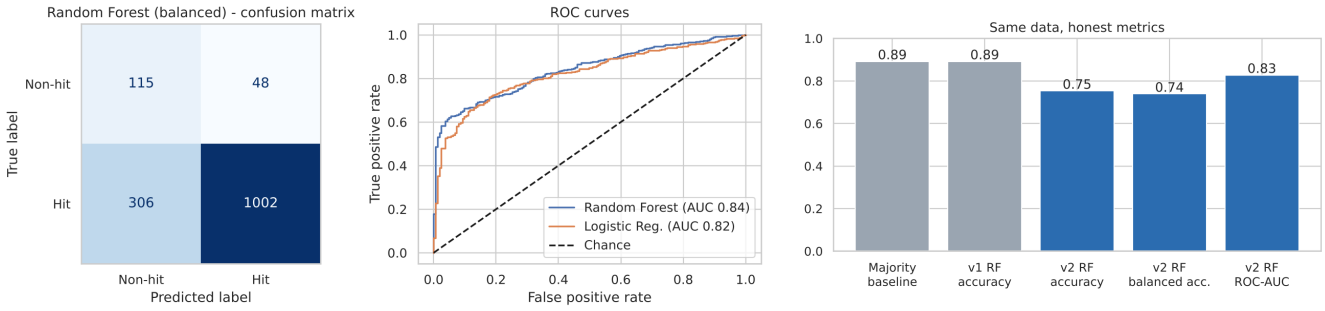


Figure 1. Left: hold-out confusion matrix and ROC curves (balanced Random Forest; 115 of 163 non-hits caught, 1,002 of 1,308 hits kept). Right: reported accuracy vs. honest metrics.

6. Conclusions

Spotify audio features carry **moderate, real signal** for separating Billboard songs from others: about 0.74–0.75 balanced accuracy and ROC-AUC 0.79–0.83, comparable to the ~75% of the Stanford paper. Random Forest ranks best by AUC and macro-F1; Logistic Regression and Decision Tree catch more non-hits (recall ≈ 0.82) at lower precision. The key lesson is methodological: accuracy alone is meaningless on 89/11 data, so baseline comparison, class weighting and imbalance-aware metrics are essential.

Limitations. (1) Non-hit labels are noisy: title-only matching can mislabel songs, and playlist songs are popular tracks, not typical ones. (2) Hits span 1946–2022 but the playlist non-hits start around 2000, so Loudness and Energy partly capture *era*. (3) Popularity is Spotify's current score, which already reflects success. (4) Random splits ignore time. Non-hit precision stays low (~0.25) because only 11% of songs are non-hits. **Future work:** a larger neutral non-hit pool (e.g. Million Song Dataset), title+artist matching, an artist-history feature, and time-based validation.

Reference. E. Georgieva, M. Suta, N. Burton, “HitPredict: Predicting Hit Songs Using Spotify Data”, Stanford CS229 Project Report, 2018. cs229.stanford.edu/proj2018/report/16.pdf